

LEGACY PROTOCOL

(LGY)

Technical Whitepaper — Revision 2026.2

Updated to clearly distinguish what is implemented on-chain today from what is planned for future development phases.

A NOTE ON HOW TO READ THIS DOCUMENT

Every section below opens with a status banner showing what is actually live in the audited smart contract today, versus what remains a planned, future capability. This revision exists specifically so that investors, partners, and the public can evaluate Legacy Protocol based on what exists right now, with a clear and honest view of the build that remains.

Table of Contents

Table of Contents.....	2
1. Abstract & Problem Statement	3
2. The Solution: Legacy Protocol (LGY)	3
3. Architecture	3
3.1 Multi-Vault IDs & Multi-Recipient Vaults	3
3.2 Decentralized Storage & Shamir's Secret Sharing	4
3.3 Private Discovery	4
3.4 Customizable Anti-Fraud Challenge Window	4
3.5 Governance Timelock & Multisig.....	4
4. Token Economics (Planned Model).....	5
4.1 Supply & Distribution	5
4.2 Activation Fee, Time-Decay Fee & Storage Surcharge.....	5
4.3 365-Day Post-Claim Recovery Window.....	5
4.4 Oracle-Pegged Dynamic Pricing.....	6
4.5 Validator/Node Collateral & Slashing.....	6
4.6 Pre-Seed SAFT & Vesting	6
5. Claims Adjudication (Planned Model).....	6
5.1 Stage One: Automated Document Review	7
5.2 Stage Two: Human Review Team	7
5.3 Stage Three: Decentralized Jury	7
6. Security Posture.....	7
7. Core Team	8
8. Roadmap.....	8

1. Abstract & Problem Statement

In the contemporary digital landscape, information custody and digital asset transmission remain exposed to central counterparty failure. Whistleblowers, journalists, executives, and crypto holders face real obstacles when trying to preserve or transfer sensitive material, legal testimony, or private wealth in the event they are silenced, incapacitated, or pass away.

This whitepaper describes both the long-term vision for Legacy Protocol and, distinctly, the subset of that vision that exists as working, reviewed code today. The problems below are the motivation for the project; the sections that follow make clear which solutions to them are real and which are still ahead of us.

1.1 The Core Problems We Are Solving For

- **The Whistleblower Vulnerability:** informants and journalists lack a trustless method to secure “insurance vaults.” If compromised, their material is often permanently lost.
- **Single-Beneficiary Bottleneck:** most digital-asset succession tools support only one static recipient, preventing simultaneous, multi-party release of sensitive material.
- **Metadata Harvesting:** centralized discovery portals can let an attacker query a target identifier and learn whether that target has anything at stake — turning the recipient into a surveillance target before any document is ever released.

2. The Solution: Legacy Protocol (LGY)

STATUS: PARTIALLY IMPLEMENTED

Legacy Protocol is a non-custodial smart contract system deployed on the BASE network (Coinbase L2), built around the idea of a multi-tenant “Dead Man’s Switch”: a user locks an encrypted reference to their material in a vault, names one or more recipients, and the vault becomes claimable either after a chosen time period elapses or after a recipient-initiated emergency claim survives a creator-controlled challenge window.

This is the layer that exists and has been security-reviewed today. The protocol's longer-term vision — described throughout the rest of this document — layers additional infrastructure (decentralized storage, a token economy, an off-chain claims-adjudication network) on top of this working core. Each later section states plainly which parts of that additional layer are built and which remain on the roadmap.

3. Architecture

3.1 Multi-Vault IDs & Multi-Recipient Vaults

STATUS: LIVE IN CONTRACT TODAY

Each vault is created with a unique ID and can name multiple beneficiaries simultaneously. A user can designate several recipients — for example, several news outlets or several family members — who can each independently claim the same vault once it unlocks.

What this means today: Live in the audited contract today, capped at 50 recipients per vault for gas-cost safety. Earlier drafts of this document described an “unbounded” recipient array — that has been corrected here to reflect the actual, tested limit.

3.2 Decentralized Storage & Shamir's Secret Sharing

STATUS: PLANNED — NOT YET BUILT

The long-term architecture calls for sharding the decryption key across a 20-of-50 Shamir's Secret Sharing scheme distributed across independent, “blind” storage nodes that have zero contextual awareness of a vault's contents or its beneficiaries.

What this means today: Not yet built. The current contract has a single configurable storage-router address (governed by the 3-of-5 multisig process described in §3.5) but does not yet implement a node network, node staking, or client-side secret sharing. Today's vaults reference an encrypted data URI; the underlying storage can be any system the creator chooses (including existing decentralized storage such as IPFS) until the dedicated node network is built.

3.3 Private Discovery

STATUS: PARTIALLY IMPLEMENTED

To prevent attackers from harvesting vault metadata, the goal is for the public-facing application to reveal nothing about which addresses have active vaults, with matches delivered privately to the correct party only.

What this means today: The contract does not expose a public function that lists vaults by recipient unless the caller already holds the correct cryptographic credential for that vault — there is no public “search by email” feature, and no leaderboard of active vaults. However, this is what we term Private Discovery rather than the stronger “Zero-Knowledge Discovery” named in earlier drafts: a sufficiently sophisticated party with direct access to contract storage (rather than the public app) could, in principle, observe vault data. True zero-knowledge discovery requires a dedicated cryptographic proof system, which is a research-stage roadmap item (see §8).

3.4 Customizable Anti-Fraud Challenge Window

STATUS: LIVE IN CONTRACT TODAY

When an emergency claim is filed against a vault, the protocol does not release it immediately. Instead, it opens a challenge window — customizable by the vault's creator between 72 hours and 30 days — during which the creator can prove they are still active and veto the claim.

What this means today: Live today. Only an address the creator explicitly designated as a recipient (verified via cryptographic signature, not a publicly readable value) can open a claim, preventing spam claims from unrelated parties. The creator can call a veto function at any time during the window to cancel a fraudulent or mistaken claim.

3.5 Governance Timelock & Multisig

STATUS: LIVE IN CONTRACT TODAY

Changes to sensitive protocol parameters — such as where data is routed, or the base pricing rate — should require both a waiting period and agreement among multiple independent parties, not a single administrator key.

What this means today: Live today: any change to the storage router or the base pricing rate requires approval from at least 3 of 5 designated core keys, plus a mandatory 14-day on-chain waiting period after the third approval before the change can take effect. No single key, including the contract deployer, can act alone or bypass the wait.

4. Token Economics (Planned Model)

STATUS: PLANNED — NOT YET BUILT

This entire section describes the planned LGY token economy. None of it exists on-chain yet: there is no LGY token contract deployed, and all fees in the current, live smart contract are denominated and paid in native ETH. This section is presented as a roadmap and a basis for discussion with investors and partners, not as a description of a live system.

4.1 Supply & Distribution

STATUS: PLANNED — NOT YET BUILT

Planned fixed supply of 1,000,000,000 LGY, with protocol fees split across node operators, a growth treasury, the founding team, and a deflationary burn.

What this means today: No LGY token contract exists yet. The fee-splitting LOGIC itself (the 80/8/7/5 proportional split) is already implemented and tested in the live contract — it currently distributes ETH fees, not LGY. Once an LGY ERC-20 contract is written, audited, and deployed, this same distribution logic can be pointed at LGY token flows.

4.2 Activation Fee, Time-Decay Fee & Storage Surcharge

STATUS: PARTIALLY IMPLEMENTED

A flat activation fee at vault creation, a continuous time-based decay fee while a vault is active, and a one-time surcharge for larger encrypted payloads, with pricing designed to stay stable against market volatility.

What this means today: The activation fee and time-decay fee both exist and are live today, denominated in ETH at a fixed rate set by protocol governance (not yet by a price oracle — see §4.4). The specific volumetric storage surcharge formula described in earlier drafts is not yet implemented as a separate on-chain calculation.

4.3 365-Day Post-Claim Recovery Window

STATUS: LIVE IN CONTRACT TODAY

Once a beneficiary successfully claims a vault for the first time, they should be able to re-download the material for free for a full year, protecting against local hardware failure.

What this means today: Live today. The contract enforces a 365-day window after first claim during which the same verified recipient can re-claim access at no additional cost.

4.4 Oracle-Pegged Dynamic Pricing

STATUS: PLANNED — NOT YET BUILT

Long-term, fees should track real-time market conditions via a price oracle, insulating users from cryptocurrency volatility.

What this means today: Not implemented, and intentionally so at this stage: LGY has not launched and has no market price for an oracle to reference. Current pricing is a fixed ETH rate, adjustable only through the same 3-of-5 multisig and 14-day timelock process described in §3.5. Once LGY has real market liquidity, pricing can be migrated to a live price feed (e.g. Chainlink) through that same governance process — no contract redesign is required when that day comes.

4.5 Validator/Node Collateral & Slashing

STATUS: PLANNED — NOT YET BUILT

Operators of storage/routing nodes would be required to lock LGY collateral (a higher tier for whistleblower-grade nodes), with misbehavior resulting in collateral being burned.

What this means today: No node registry, staking, or slashing contract exists yet. This depends on both the LGY token (§4.1) and the decentralized storage network (§3.2) being built first.

4.6 Pre-Seed SAFT & Vesting

STATUS: PLANNED — NOT YET BUILT

Pre-seed capital is intended to be raised via a standard SAFT structure with a 12-month cliff and 24-month linear vesting.

What this means today: This is a legal and financial structure, not smart-contract logic that exists today. A vesting contract is straightforward to deploy once the LGY token contract exists, using well-established, auditable patterns; the SAFT terms themselves are a legal agreement between the company and investors, independent of the contract code.

5. Claims Adjudication (Planned Model)

STATUS: PLANNED — NOT YET BUILT

This section describes the planned multi-stage process for verifying emergency claims (e.g. death certificates) at scale. None of the adjudication logic below exists on-chain today. What the live contract DOES already support is the final step common to all three planned stages: once a claim is

determined valid by whatever process is used, a designated recipient can open and, if uncontested through the challenge window, complete the claim on-chain using a verified cryptographic signature.

5.1 Stage One: Automated Document Review

STATUS: PLANNED — NOT YET BUILT

Digital death certificates or registry QR codes would be parsed by an automated document-reading process that authenticates against sovereign registries directly.

What this means today: Not implemented. This is an off-chain document-processing service that does not exist yet. Once built, it would interface with the live contract only at the final step — confirming a claim that the contract already knows how to process.

5.2 Stage Two: Human Review Team

STATUS: PLANNED — NOT YET BUILT

Complex or low-confidence certificates would be routed to a small team of human reviewers who verify the registry record and post the result for peer confirmation.

What this means today: Not implemented. An off-chain workflow and review tooling would need to be built; the on-chain footprint would again be limited to confirming the final outcome.

5.3 Stage Three: Decentralized Jury

STATUS: PLANNED — NOT YET BUILT

At high transaction volume, a decentralized, collateral-backed jury pool would vote anonymously on contested claims, with dishonest jurors losing their stake.

What this means today: Not implemented. This is a substantial, standalone subsystem (collateral, anonymous voting, slashing) comparable in scope to existing decentralized-court protocols. It is sequenced as a later-phase roadmap item, not a near-term deliverable.

6. Security Posture

STATUS: LIVE IN CONTRACT TODAY

The live smart contract has been through an internal security review pass covering, among other things, reentrancy protection, front-running resistance on claim authorization (using cryptographic signatures rather than exposing secrets in public transactions), denial-of-service resistance on unbounded loops, and a real on-chain multisig-plus-timelock for sensitive parameter changes.

This internal review does not substitute for an independent third-party audit. A formal audit (e.g. via a firm such as CertiK or Hacken) is budgeted into the planned use of pre-seed funds and is treated as a prerequisite — not an optional step — before any user funds interact with the contract on a public mainnet.

7. Core Team

- Founder & CEO: Tomas Petrauskas — corporate operations, regulatory strategy, and venture execution.
- Co-Founder & CTO: AI Systems Partner — smart contract architecture and protocol engineering.

8. Roadmap

Sequenced by technical dependency rather than by calendar ambition — each phase requires the one(s) before it.

Phase	Scope	Depends On
Phase 0	Contract hardening: independent third-party audit, formal compiler verification, migration of hand-rolled cryptography to audited libraries.	Completed internal review (this document)
Phase 1	Public testnet launch of the core vault product (create, extend, claim, veto, cancel) with ETH-denominated fees and the off-chain notification relay built out.	Phase 0
Phase 2	LGY ERC-20 token contract, vesting contract for SAFT terms, migration of fee payments from ETH to LGY.	Phase 1 live and stable
Phase 3	Oracle-based dynamic pricing once LGY has real market liquidity.	Phase 2
Phase 4	Decentralized storage network and/or integration, client-side Shamir's Secret Sharing, node staking and collateral.	Phase 2 (token economics)
Phase 5	Claims adjudication system: automated document review, human review team, decentralized jury.	Phase 1 (claims flow proven in production)
Phase 6	True zero-knowledge discovery (cryptographic membership proofs) replacing today's Private Discovery model.	Research-stage; no fixed dependency

A detailed, line-by-line cross-reference of every claim in this document against the audited contract is available as a separate companion document (Whitepaper vs. Code Gap Analysis) for technical due diligence.